



End-to-End E-Commerce Sales Analysis

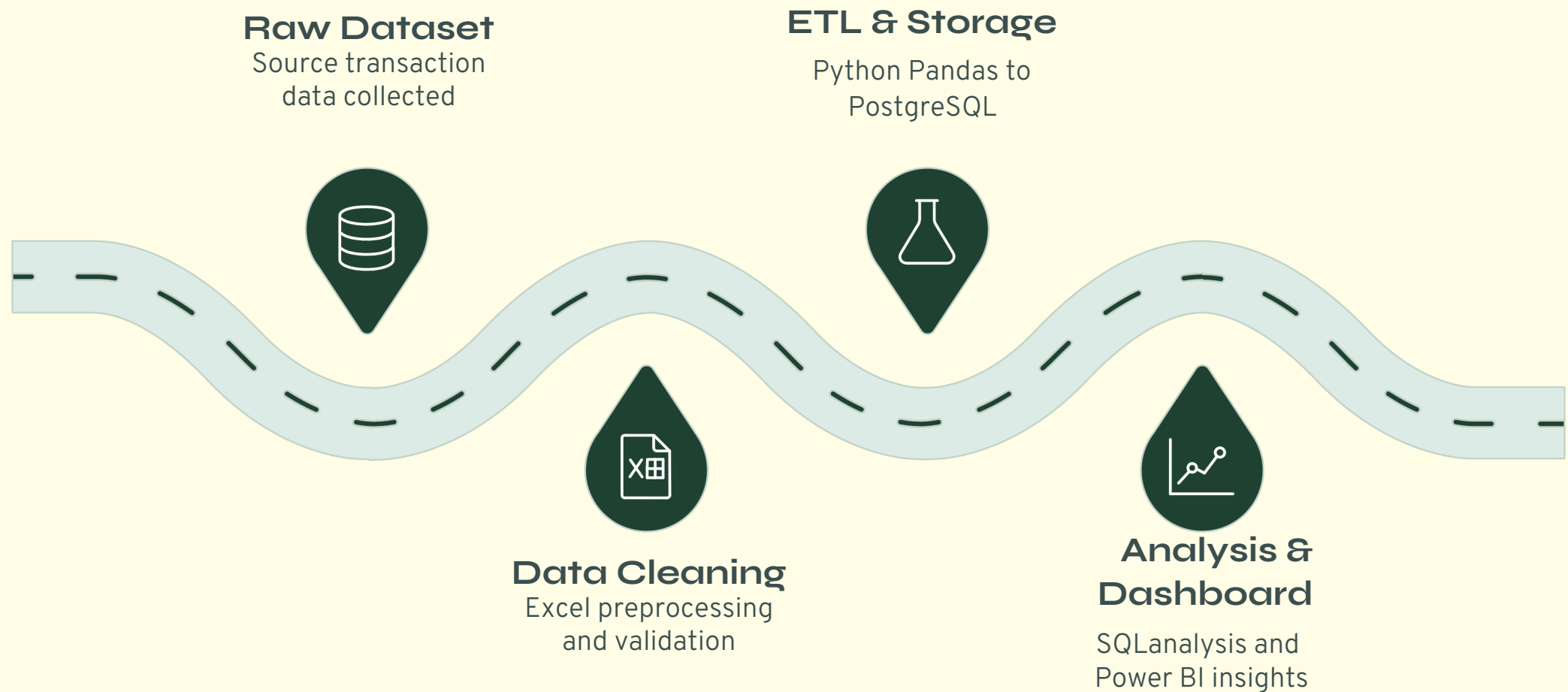
From RawData to Interactive Business Dashboard

Tech Stack: Microsoft Excel • Python • PostgreSQL • SQL • Power BI

Author: Nitin Singh

Project Overview & Workflow

Objective: Deliver an end-to-end analytics pipeline that transforms raw transaction data into actionable business insights via a reproducible ETL, analysis, and an interactive Power BI dashboard for stakeholders.



Pipeline highlights: repeatable cleaning, programmatic EDA, relational storage for scale, SQL-driven metrics, and an interactive BI layer for exploration and decisions.

Step 1 — Data Cleaning

Primary Actions

- Removed duplicate records
- Checked & imputed missing values
- Standardized column names and formats
- Corrected data types (dates, numerics)
- Exported cleaned dataset (CSV)

Tools

Microsoft Excel for initial profiling, quick fixes, and sample exports.

DATA CLEANING & PREPARATION

to clean, consistent and analysis-ready dataset

DATA)				
	E	F	F	II
	category	Sales	Profit	Region
pier	Technology	261.9500	41.9136	West
mb Bind	Office Supplies	731.9400	219.5820	West
mb Bind	Office supplies	14.6200	6.8714	West
ar Table	Furniture	957.5775	-383.0310	East
	Office Supplies	22.3680	2.5164	East
	Technology	48.8500	14.1694	East
	Technology	7.2800	1.9658	East
	Technology	907.1520	-54.4291	East
	Technology		-695.2600	East
	Office Supplies	18.5040	5.7825	Central
	Technology	114.9000	34.4700	Central
	Office Supplies	1706.1840	85.3092	Central
	Technology	911.4240	68.3568	Central
	Office Supplies	15.5520	5.4432	Central



AFTER CLEAN				
	A	B	C	
	order_id	order_date	customer_name	
1	CA-2014-152156	2014-11-08	Claire Gule	Canon ImageCL
2	CA-2014-152156	2014-11-08	Claire Gule	Fellowes PB500
3	CA-2014-138688	2014-06-12	Darrin Van Huff	Fellowes PB500
4	US-2015-108966	2015-10-11	Sean O'Donnell	Bretford CR4500
5	US-2015-108966	2015-10-11	Sean O'Donnell	BOXI File Box, L
6	CA-2014-115812	2014-06-09	Brosina Hoffman	Xerox 1967
7	CA-2014-115812	2014-06-09	Brosina Hoffman	Holmes HEPA A
8	CA-2014-115812	2014-06-09	Brosina Hoffman	Xerox 1967
9	CA-2014-115812	2014-06-09	Brosina Hoffman	Xerox 1967
10	CA-2014-167164	2014-11-24	Andrew Allen	Newell 321
11	CA-2014-167164	2014-11-24	Andrew Allen	Mitel 5320 IP P
12	CA-2014-143336	2014-08-27	Irene Maddox	Newell 321
13	CA-2014-143336	2014-08-27	Irene Maddox	Mitel 5320 IP P
14	CA-2014-143336	2014-08-27	Irene Maddox	Newell 321

PERFORMED

- 05 Standardized Column Names**
Converted all column names to lowercase with underscores
- 06 Corrected Data Inconsistencies**
Fixed typos, inconsistent categories and text values
- 07 Validated Data**
Checked for outliers and invalid entries
- 08 Saved Cleaned Dataset**
Exported cleaned data as CSV for further analysis

KEY IMPROVEMENTS

- ✓ Removed duplicate rows
- ✓ Standardized date format (YYYY-MM-DD)
- ✓ Consistent text formatting
- ✓ Uniform category names
- ✓ Removed leading/trailing spaces
- ✓ Corrected typos and inconsistencies
- ✓ Proper data types for accurate analysis
- ✓ Clean, reliable and analysis-ready data

ed into a clean, consistent and reliable dataset
is and Power BI dashboarding.

 **Output File**
cleaned_dataset.csv

Next St
Python E

Step 2 – Exploratory Data Analysis (Python / Pandas)



Pipeline in Python

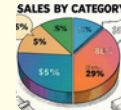
- Load cleaned CSV
- dtype conversions
- summary statistics
- time series
- resampling
- groupby
- aggregations.



**Monthly Sales
Trend**



**Monthly
Profit Trend**



**Category &
Region
Analysis**



**Top & Loss-
Making
Products**

Step 3 – Database & SQL Analysis (PostgreSQL)

6

**Step 4 –
Business Analysis
using SQL**



```
-- Total Sales
SELECT ROUND(SUM(sales), 2) AS total_sales
FROM orders;

-- Total Profit
SELECT ROUND(SUM(profit), 2) AS total_profit
FROM orders;

-- Top 5 Products by Sales
SELECT product_name, ROUND(SUM(sales),2) AS total_sales
FROM orders
GROUP BY product_name
ORDER BY total_sales DESC
LIMIT 5;
```

Action: Loaded cleaned CSV into PostgreSQL for reliable storage, joins, and performant aggregations. Structured schema supports repeatable analysis and team access.



Representative SQL Queries

- Total Sales, Total Profit
- Average Order Value
- Top Products & Top Customers
- Sales by Region / Category
- Profit margin, GROUP BY, HAVING, ORDER BY, LIMIT



Reproducibility

Versioned SQL scripts and schema migrations stored on GitHub. Placeholder for SQL screenshots and example queries.

Step 4 – Dashboard Development (Power BI)

Goal: Build an interactive dashboard for executives and analysts to explore sales, profit, product, and regional performance with filter-driven insights.

Dashboard features: KPI cards, Monthly Sales Trend, Monthly Profit Trend, Top 5 Products, Sales by Category, Interactive Filters, Region & Category analysis.



Interactive Dashboard Features



Year Filter

Quickly switch fiscal years to compare trends and seasonality.



Category Filter

Drill into product categories to reveal high-margin segments.



Region Filter

Isolate geographic performance and regional KPIs.



Dynamic KPI Cards

KPI cards recalc based on filters and drive quick status checks.



Key Business Insights

2M
Total Sales

Total Revenue across the period

286K
Total Profit

Aggregate profit after cost calculations.

5K
Orders

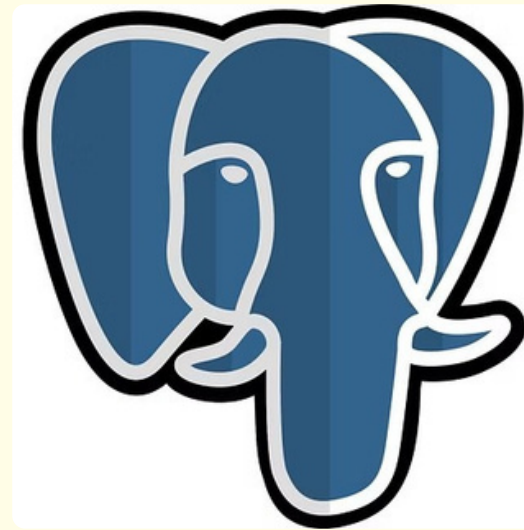
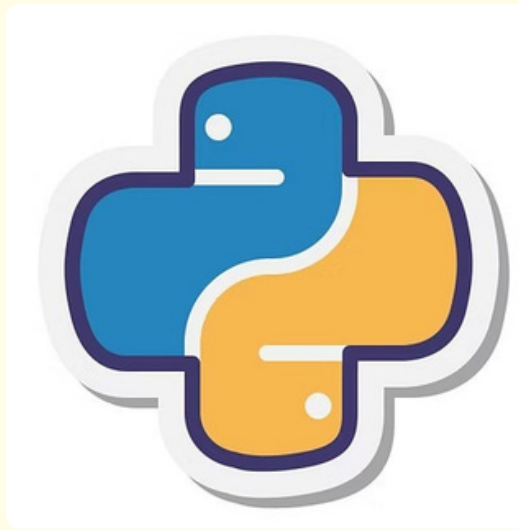
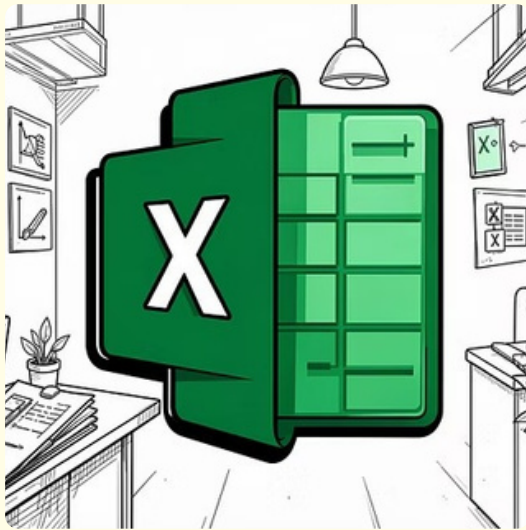
Total number of unique orders processed.

12.47%
Profit Margin

Overall margin across sales (Profit ÷ Sales).

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- **Top category:** Technology generated the highest sales volume and revenue share.
 - **Peak month:** November recorded the highest monthly sales – useful for promotion planning.
 - **Top brand:** Canon products drove the largest single-brand revenue.
 - **Impact:** Interactive dashboard enables data-driven decisions for pricing, inventory, and promotions.

Tools & Skills Used



Core Skills

- Data Cleaning & Transformation
- Exploratory Data Analysis (Pandas)
- Database Design & Management (PostgreSQL)
- Advanced SQL Querying
- Dashboard Design & Business Intelligence
- Data Visualization & Storytelling

Engineering Practices

- Version control (GitHub)
- Reproducible scripts & documentation
- Modular SQL and ETL workflows

Project Summary & Next Steps

This project demonstrates a complete end-to-end analytics workflow: raw data ingestion, cleaning, programmatic EDA, structured storage, SQL analysis, and an interactive Power BI dashboard that surfaces actionable business insights.

Key Takeaways

- Data Cleaning establishes trust in downstream metrics
- EDA revealed seasonality and product-level performance
- Relational DB enables scalable, auditable queries
- Power BI dashboard enables cross-filtering and rapid insight

Recommended Next Steps

1. Embed dashboard in stakeholder portal and collect feedback
2. Automate ETL: schedule cleaning & DB loads
3. Expand KPIs: cohort retention, LTV, promotion lift
4. Validate with finance for forecasting & planning

Thank you

Created by Nitin Singh

<https://github.com/nsr-dev-in/E-Commerce-Sales-Analysis>